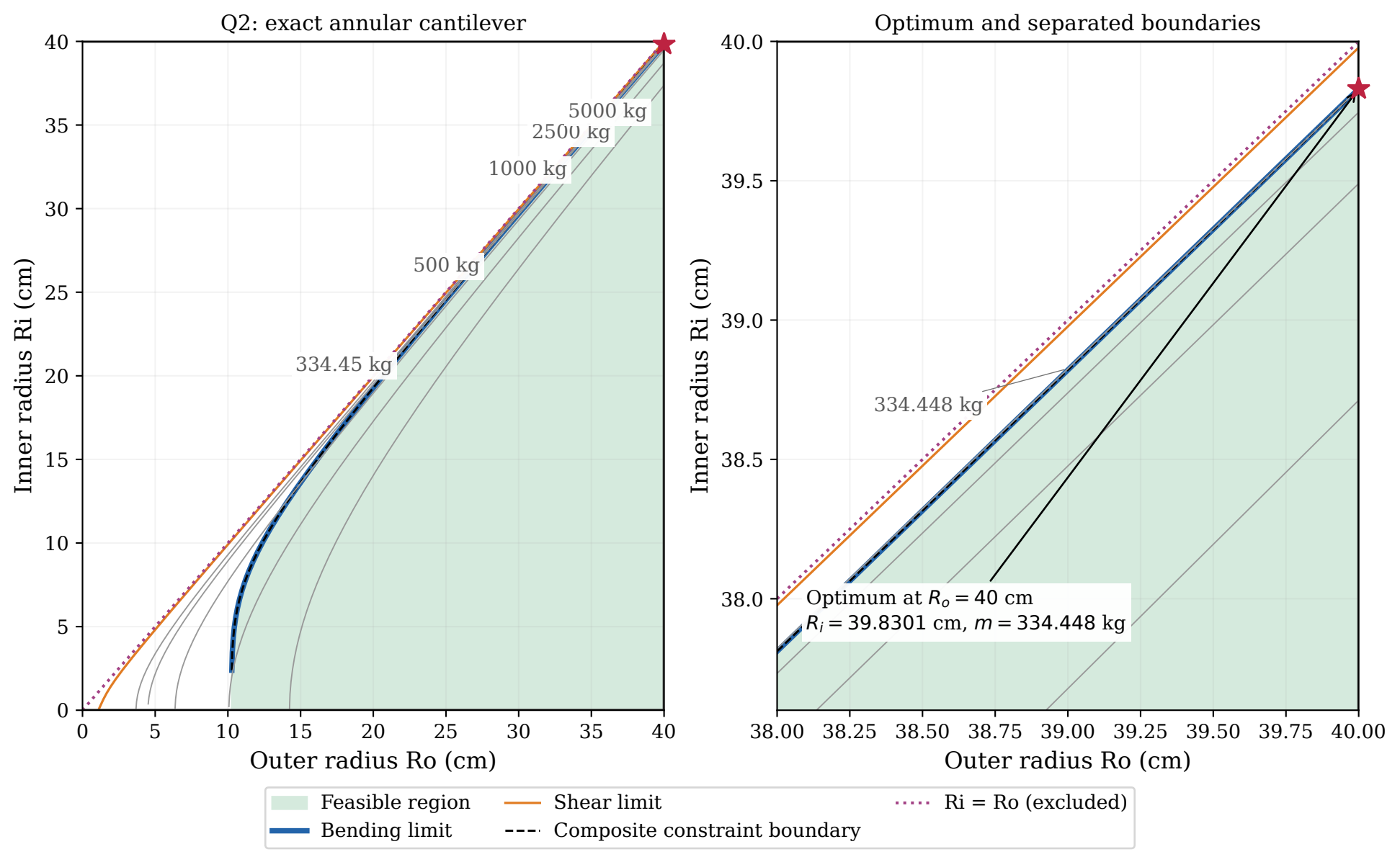

Q2: exact annular cantilever
Optimum and separated boundaries
Inner radius Ri (cm)
Outer radius Ro (cm)
5000 kg
2500 kg
1000 kg
500 kg
334.45 kg
334.448 kg
Optimum at $R_o = 40$ cm
$R_i = 39.8301$ cm, $m = 334.448$ kg
Feasible region
Bending limit
Shear limit
Composite constraint boundary
Ri = Ro (excluded)